

Pretrained protograph codes
one vector per class & relation

① **Normalize each code to norm 8**

$$c \leftarrow 8c/\|c\|$$

remove pretrain-frequency bias

② **Unit-normalize
each component,
then sum**

bind features; counts $\rightarrow$ length

③ **Global rescale**

$$\text{mean norm} \rightarrow 8$$

protect the init (saturate σ)

④ **Cap row norms at 16**

DBpedia only

**only on DBpedia (real graphs):
unfreeze hub entities**

Protected fine-tune